

Daniele Battesimo Provenzano

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PROFESSIONAL EXPERIENCE

Forensic Department of the Italian Police

Technical Inspector in Forensic Science

Rome, Italy

Sept. 2025 – present

- Performing Scanning Electron Microscope (SEM) and microanalytical investigations within the Gunshot Residue Analysis Division.
- Teaching in specialized training courses for forensic science trainees, focusing on experimental techniques, analytical procedures, and evidence evaluation.

“Italian Association for the Teaching of Physics” (AIF) Various locations across Italy

Teacher and Trainer

May 2019 – present

- Design and deliver lectures, laboratory activities, and problem-solving sessions for students preparing for the Italian and International Physics Olympiads.
- Develop examination materials and contribute to national programs aimed at improving physics education and fostering advanced student training across Italy.

EDUCATION

Scuola Normale Superiore

Ph.D. in Physics: Plasmonics and Metamaterials; ABD

Pisa, Italy

Nov. 2020 – Sept. 2025

Diploma (M.Sc.) in Physics of Matter; 100/100 cum Laude

Oct. 2018 – Sept. 2020

University of Pisa

M.Phil. in Condensed Matter Physics; 110/110 cum Laude

Pisa, Italy

Sept. 2018 – Oct. 2020

Scuola Superiore di Catania

Diploma (M.Sc.) in Physics; 70/70 cum Laude

Catania, Italy

Oct. 2015 – Sept. 2018

University of Catania

B.Sc. in Physics; 110/110 cum Laude

Catania, Italy

Oct. 2015 – July 2018

“Liceo Scientifico Leonardo da Vinci”

High School Diploma; 100/100 cum Laude

Reggio Calabria, Italy

Sept. 2010 – June 2015

ADDITIONAL COURSES

“GSR Analysis”: Four-week intensive training program on the use of Scanning Electron Microscopy (SEM) and the application of X-ray, Raman, and FTIR spectroscopies for the characterization of organic and inorganic gunshot residues (Oct.–Nov. 2025).

School for Forensic Police Inspectors: Selected through a national open competitive examination and completed a six-month training program in forensic science during Ph.D. studies. The program included Italian and international law, as well as experimental techniques used by the *Forensic Department of the Italian Police*, with a specialization in ballistics and gunshot residue analysis. Successfully appointed as *Technical Inspector* upon completion of the training. (June 2023 – Gen. 2024)

“24 ECTS for Teaching”: Completed four prerequisite courses in Pedagogy, Psychology, Anthropology, and Teaching Methods, required for teacher qualification in Italy. (Summer 2022)

“English for Scientific Communication”: Completed a course on effective scientific writing and presentation, with a focus on preparing research papers and delivering talks at international conferences. (Spring 2022)

Afternoon English Courses: Attended continuous English language courses at a British School in my hometown, focusing on grammar, writing, and conversation practice (4 hours per week). Earned Cambridge English certifications up to level C1. (Oct. 2009 – June 2015)

TEACHING EXPERIENCE

Scuola Normale Superiore

Pisa, Italy

Lecturer, Teaching Assistant and Examiner

- **“Physics School in Pisa”**: Prepared and gave 28 hours of lectures about topics in Classical and Modern Physics and led the relative problem-solving sessions, examined and graded 200 applicants per year through both written and oral exams. (2019 – 2025)
- **“University Orientation Courses”**: Designed and delivered 12 hours of interactive lessons on Problem-Solving Techniques, aimed at 260 handpicked high school students. (June - July 2024)
- **“Classical Mechanics and Thermodynamics” course**: Served as a teaching assistant, delivering 30 hours of problem-solving sessions for first-year university students. (Fall 2020 – Spring 2021)

“Italian Association for the Teaching of Physics” (AIF) Various locations across Italy

Lecturer, Tutor and Examiner

- **“One step away from IPhO”**: Prepared and gave 32 hours of lectures on Celestial Mechanics, Electromagnetism and Thermodynamics, aimed at the top 15 Italian students preparing to compete in the International Physics Olympiad. Selected the team members taking part in IPhO and EuPhO through theoretical and experimental examinations. (May 2019/23/24/25)
- **Physics Olympiad Summer School**: Led 16 hours of problem-solving sessions on Orbital Mechanics, Estimation Problems, Dimensional Analysis and Fluid Dynamics, aimed at 20 handpicked students taking part in the Physics Olympiad. (Aug. 2019/24/25)

Scuola Superiore di Catania

Catania, Italy

Lecturer, Tutor and Examiner

- **“Training camp for the Physics Olympiad”**: Prepared and gave 6 lectures about Electrostatics, Problem-Solving Techniques and Fluid Dynamics, aimed at 20 handpicked students taking part in the national finals of the Physics Olympiad. Led the relative problem-solving sessions, prepared and graded the final theoretical competition. (Mar. 2023/24/25)
- **“Training camp for the Math Olympiad”**: Led two problem-solving sessions focused on Combinatorics and Game Theory, aimed at 25 handpicked students taking part in the national finals of the Math Olympiad. (April 2016 and 2017)

PUBLICATIONS

D. B. Provenzano, Z. Bacciconi, I. Carusotto, M. Dalmonte, G. C. la Rocca, *Graviton-Polaritons in FQH systems coupled to optical cavities*, in preparation.

Daniele B. Provenzano, *A comprehensive tutorial on Bernoulli’s equation*, in preparation.

Daniele B. Provenzano, *The incredible amount of physics behind the Heron’s fountain*, under review by the AJP.

Daniele B. Provenzano, *Debunking Boyle’s self-flowing flask*, under review by the EJP.

Stefania Carletti and Daniele B. Provenzano, *The Ninth Edition of the European Physics Olympiad*, chapter of *Speciale Olimpiadi 2025* (in Italian), published by Associazione per l’Insegnamento della Fisica (AIF). (Coming in Dec. 2025)

Daniele B. Provenzano, Andrea Stefanini, [Unblowing bubbles: understanding the physics of bubble deflation through a straw](#), Am. J. Phys. **93**, 797–805 (2025)

Daniele B. Provenzano and Giuseppe C. la Rocca, [Bulk and surface modes in a one-dimensional gyro-uniaxial photonic crystal](#), Phys. Rev. A **110**, 033514 (2024).

Daniele B. Provenzano and Giuseppe C. la Rocca, [Interface mode between gyroelectric and hyperbolic media](#), J. Opt. Soc. Am. B **40**, 172 (2023).

CONFERENCE TALKS AND POSTERS

Metamaterials'2025, Amsterdam (NL), Sept. 2025

Graviton-Polaritons in FQH Systems Coupled to Optical Cavities (P)

GIREP-EPEC Conference, Leiden (NL), July 2025

When Assumptions Make the Difference: The Curious Case of the Mistreated Bernoulli's Equation (T)

META 2024, Toyama (JP), July 2024

Photonic Modes in Gyrotropic-Hyperbolic Heterostructures (P)

AWARDS & ACHIEVEMENTS

Italy Team Leader at EuPhO 2025: Coached and led the Italian team at the European Physics Olympiad, held in Sofia, Bulgaria. (June 2025)

Best student of the Technical Ballistics Inspector course: after six months of training, I ranked first in the final examinations among the ballistics inspector trainees and third overall among all the forensic science trainees, and was awarded a bronze medal. (Dec. 2023)

First Place in the National Open Competitive Exam for Technical Ballistic Inspectors: Ranked first in the decennial national exam for ballistic experts to enroll in the Forensic Department of the Italian Police. (Apr. 2023)

Gold Medal – 53rd Rudolf Ortway International Competition in Physics. (Feb. 2023)

Absolute Winner – 50th Rudolf Ortway International Competition in Physics. (Dec. 2019)

SNS Scholarship: Awarded full scholarship (including room, board and monthly stipend) after passing the selection process to enroll at Scuola Normale Superiore as a Master's student. (Oct. 2018 – Sept. 2020)

SSC Scholarship: Granted full scholarship (room and board) after passing the selection process to enroll at Scuola Superiore di Catania as an undergraduate student. (Oct. 2015 – Sept. 2018)

PROJECTS RELATED TO PHYSICS EDUCATION

Italian Physics Olympiad (2019 – present) | [Italian website](#)

- Organized since 1987 by the *Italian Association for the Teaching of Physics* (AIF), the Italian Physics Olympiad selects and trains the country's top students for international competitions from among approximately thirty thousand participants each year. I joined the national training team in 2019. Since then, students I have trained have earned 2 gold, 16 silver and 28 bronze medals at IPhO and EuPhO. In recognition of my contributions, I was appointed to the national committee responsible for preparing theoretical and experimental problems and for evaluating contestants at each stage of the competition.

Physics School in Pisa (2019 – present) | [Italian website](#)

- Since 2018, a team of physics students of Scuola Normale Superiore (SNS) has organized an annual one-week school for 24 top-performing Italian high school students. The program combines advanced problem solving sessions with an introduction to university-level physics through both classroom lectures and laboratory activities. In 2019, the project won the “Didactics of Physics” award from the Italian Physical Society. Since enrolling at SNS in 2019, my main duties have included evaluating and grading about 200 applicants per year through written and oral examinations, preparing original exam problems, and delivering lectures and problem-solving sessions.

Team Physics Competition (2023 – present) | [Italian website](#)

- The TPC is multi-stage event for Italian high school students that involves approximately three thousand participants each year, created by my university peers and me. My primary responsibility involves creating, refining and selecting the theoretical problems used in the competition.

Danny & Tony's Lab (2025 – present)

- Together with a computer scientist friend, I am developing a fully custom-built virtual and interactive physics laboratory hosted on a website. The platform will allow users to perform original physics experiments (some inspired by my published educational papers) in a 3D environment where they can walk around and interact with the apparatus. One section of the lab will include an astronomical observatory for celestial mechanics simulations. The project aims to serve as an innovative and highly engaging tool for physics education.

“Problem-Solving Techniques in Physics, with Original Problems and Debunked Myths”

- This is a textbook on problem-solving techniques that I am writing in my free time, aimed at students preparing for the Physics Olympiad and first-year university students. It is specifically designed to debunk common myths and address misconceptions often found in traditional teaching and textbooks. I aim to complete it by 2027.

CERTIFICATIONS AND SKILLS

Languages: English (C1, 195/210, certificated by Cambridge English), French (B1, certificated by Scuola Normale Superiore).

Major computer skills: MATLAB, Mathematica, Python, $\text{\LaTeX}$, Git, HTML, video editing SWs.

Others: Fire Safety and First Aid (both certificated by the Italian Police).